# ERIC DECASPER

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## EDUCATION

**California Polytechnic State University** | San Luis Obispo, CA **Expected December 2026**
B.S. Mechanical Engineering
**Coursework:** Mechanical Vibrations, Heat Transfer, Controls I & II, Fluid Mechanics I & II, Machine Learning in ME

## SKILLS

**CAD and analysis:** SolidWorks, Onshape, Fusion 360, GD&T, tolerance stack-up analysis, DFM, FEA (ANSYS)
**Fabrication:** Mill and lathe, wire-EDM tooling, 3D printing (FDM, SLA, SLS)
**Programming:** Python, MATLAB, C++ (Teensy firmware), Java
**Languages:** English, Swiss German (conversational), French (conversational)

## EXPERIENCE

**Western Magnetics Company** | South San Francisco, CA **June 2026 - August 2026**
*Mechanical Engineering Intern*

- Built a quick-release, vibration-isolated pneumatic fixture that let the end-of-line tester reliably grade every production motor in a 45-second cycle and made fast design studies on vibration and drag possible; 4,400+ cycles by August.
- Traced motor KV variation to magnet thickness and lamination tolerance stack-up, then built a Keyence measurement fixture for stators; the team used the data to adjust winding programs and bring KV back into range.
- Designed servo-press shear tooling from tungsten carbide with resistive heaters to test magnet adhesives across a range of temperatures, validated to over 5,000 N of press force.
- Used the press to qualify an adhesive 2-3x stronger than the incumbent, cutting adhesion failures from over 70% to near zero, and set up an in-line bond-strength test that caught an under-cured batch before it shipped.
- Created tooling for several automated stations, including custom wire-EDM tooling owned from design through fabrication, and set up an outside machining partner with about a one-week turnaround on urgent tooling.
- Built the company's first drone platform for flight-testing its motors and collecting in-flight performance data.

**Voliro AG** | Zurich, Switzerland **July 2025 - December 2025**
*Mechanical Engineering Intern*

- Prototyped new camera and radar systems by integrating fisheye cameras and radars into the drone's main body components, running field-of-view studies to ensure full coverage around the drone, and routing cables with strain relief.
- Designed and validated a cooling solution for a next-generation drone radio, building a 40°C environmental chamber to verify sustained operation at worst-case ambient temperature.
- Revived a drone spot-repair capability (surface prep and paint) that an earlier team had dropped at proof of concept: built the first flight-worthy prototype in one week, then owned mechanical design and testing of both payloads with MontiPower through flight tests. Voliro now sells it as a service.
- Took the wire-wheel sanding payload from a PLA prototype to CNC aluminum and PA12 parts.

## PROJECTS

**NDT Scanning Arm, Voliro-sponsored senior project** | Cal Poly SLO **January 2026 - Present**

- Leading a 4-person team building a drone-mounted arm that sweeps an EMAT ultrasonic sensor across steel structures to find defects at height.
- Designed the two-link carbon fiber arm under a 1 kg limit, then built and programmed its control system: a Teensy 4.1 running inverse kinematics on two Dynamixel XM430 servos, the first Voliro payload with its own microcontroller.
- Bench-tested on real steel assets; now integrating it onto a Voliro drone for flight testing at Cal Poly.

**Aerial Manipulation with Reinforcement Learning** | Cal Poly SLO **September 2026 - Present**

- Training reinforcement-learning controllers in Isaac Lab toward a learned controller for the senior-project arm.

## LEADERSHIP

**FRC Team 670, Homestead High School** | Mechanical Lead **August 2021 - June 2022**

- Added a custom vacuum and compressed-air system to a CNC router, giving the team its first in-house aluminum cutting; FIRST Quality Award and 2022 Cal Games champions.